Photoelectric Effect

- Einstein considered the possibility if

  Max Planck's hypothesis was right about the energy quanta [EM radiation in cavity walls].

- If so, then the hypothetical framework could also explain the energy absorption from EM radiation by the surface of the photoelectrode.



Photoelectric Effect

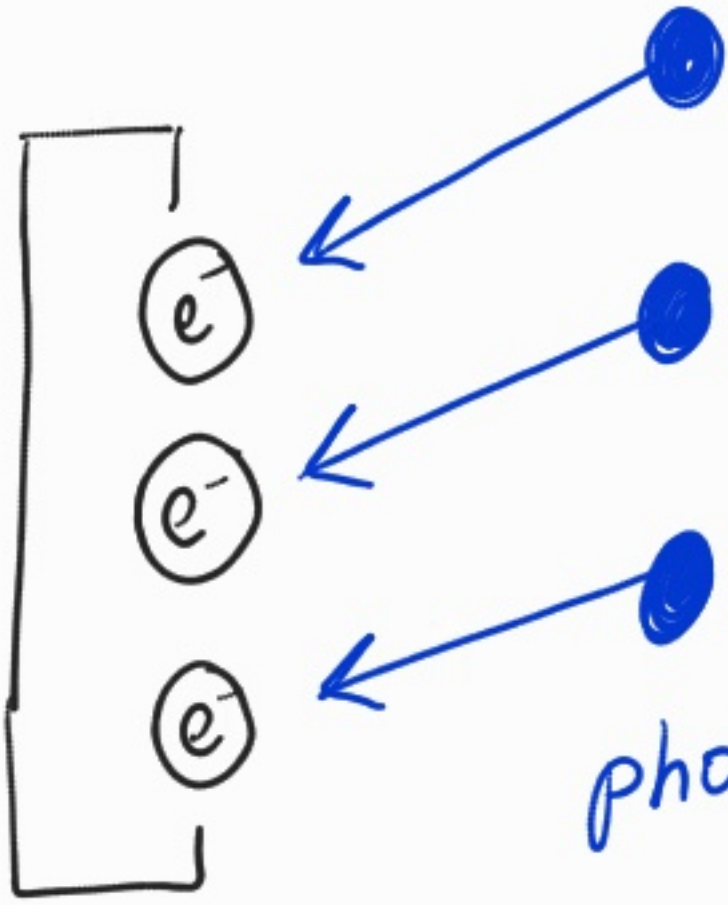
Einstein's Approach

- a beam of monochromatic light with frequency, f is made of photons.
- a photon is a particle of light & it moves at the speed of light.
- photons carry an energy quantum of E_f where its energy only depends on its frequency, f:

$$E_f = hf$$

h - Planck's Constant

Photoelectric Effect



photon

surface photoelectrons

"All or nothing" energy transfer

the photon ceases to exist after transfer.

$$E_f = K_{max} + \phi$$

$E_f \Rightarrow$ Photon energy

$K_{max} \Rightarrow$ Kinetic energy

$\phi \Rightarrow$ energy required to detach photoelectron from surface

Photoelectric Effect

ϕ (eV) $\Rightarrow$ detachment energy is also called the work function.

Work Function for some metals.

metal	ϕ (eV)
Ag	4.73
Al	4.08
Pt	6.35

- To determine the Kinetic energy (max) of photoelectrons:

$$E_f = hf$$

$$K_{max} = E_f - \phi$$

$$= (hf - \phi)$$

Photoelectric Effect

Einstein's Intrepretation:

i/ Absence of lag time

- interaction are between

individual electrons ⇔ individual photons

INSTANTANEOUS

ii/ EM intensity & KE of electrons

- interaction time cannot be increased by lowering EM intensity.
- EM intensity → no. of photons aiming at metal surface per unit time

Con't →

Photoelectric Effect

iii/ Existence of cut-off frequency (f_c)

- This frequency follows:

$$K_{max} = hf - \phi$$

$\Rightarrow$ some threshold frequency where $K_{max} = 0$:

$$\therefore hf_c - \phi = 0 \rightarrow \boxed{f_c = \frac{\phi}{h}}$$

- Cut-off frequency only depends on ϕ of the metal. $\rightarrow \uparrow \phi \rightarrow \uparrow f \rightarrow \uparrow E$
- Photon with energy lower than $E_c = hf_c$ do produce the photoelectric effect.

Photoelectric Effect

Wavelength analysis :

$$\lambda f = c \quad \text{(for EM waves)}$$

$\Rightarrow$ determine the cut-off wavelength.

$$\lambda_c = \frac{c}{f_c}$$

$$\rightarrow f_c = \phi / h$$

$$\lambda_c = \frac{c}{\phi/h} = \left[\frac{hc}{\phi}\right]$$

$\therefore$ when wavelength of the incident EM radiation, $\lambda > \lambda_c$, the photoelectric effect does not occur.

Photon Momentum

- As energy particles, photons are massless.
- As with all types of propagation of pure energy (i.e. EM waves), photons have only one speed when they travel ⇒ speed of light, c.

How is it possible to determine the momentum of photon when it does not have mass

?

Photon Momentum

- Relation from the special theory of relativity.
- for particles/objects with mass moving at velocities close to the speed of light ($v \to c$).

Relativistic Energy

$$E^2 = p^2c^2 + m_0^2c^4$$

p - linear momentum

m_0 - rest mass

$c = 3 \times 10^8$ m/s

Photon Momentum

→ Apply relativistic energy relation to a photon:

Since the photon has no mass → $m_o = 0$

$$\therefore \quad \underset{f}{E} = p_f c \qquad \left[\begin{matrix}\text{photon}\\ \text{energy}\end{matrix}\right] \qquad \text{——— (1)}$$

→ Planck's Energy Quanta:

$$\underset{f}{E} = hf \quad \& \quad f = c/\lambda$$

$$\therefore \quad E_f = \frac{hc}{\lambda} \qquad \text{——————— (2)}$$

$$p_f c = \frac{hc}{\lambda} \rightarrow \therefore \boxed{p_f = \frac{h}{\lambda}}$$

↳ Photon momentum

Photon Momentum

- Photon's momentum as a vector :

$$P_f = h/\lambda$$

→ Recall → wave number : $k = \frac{2\pi}{\lambda}$

$$P_f = \frac{h}{2\pi/k} = \frac{h}{2\pi} \cdot k$$

→ Reduced planck's constant $\hbar = \frac{h}{2\pi}$

$$P_f = \hbar k$$

→ Introduce wave vector : $k = |\vec{k}| = \frac{2\pi}{\lambda}$

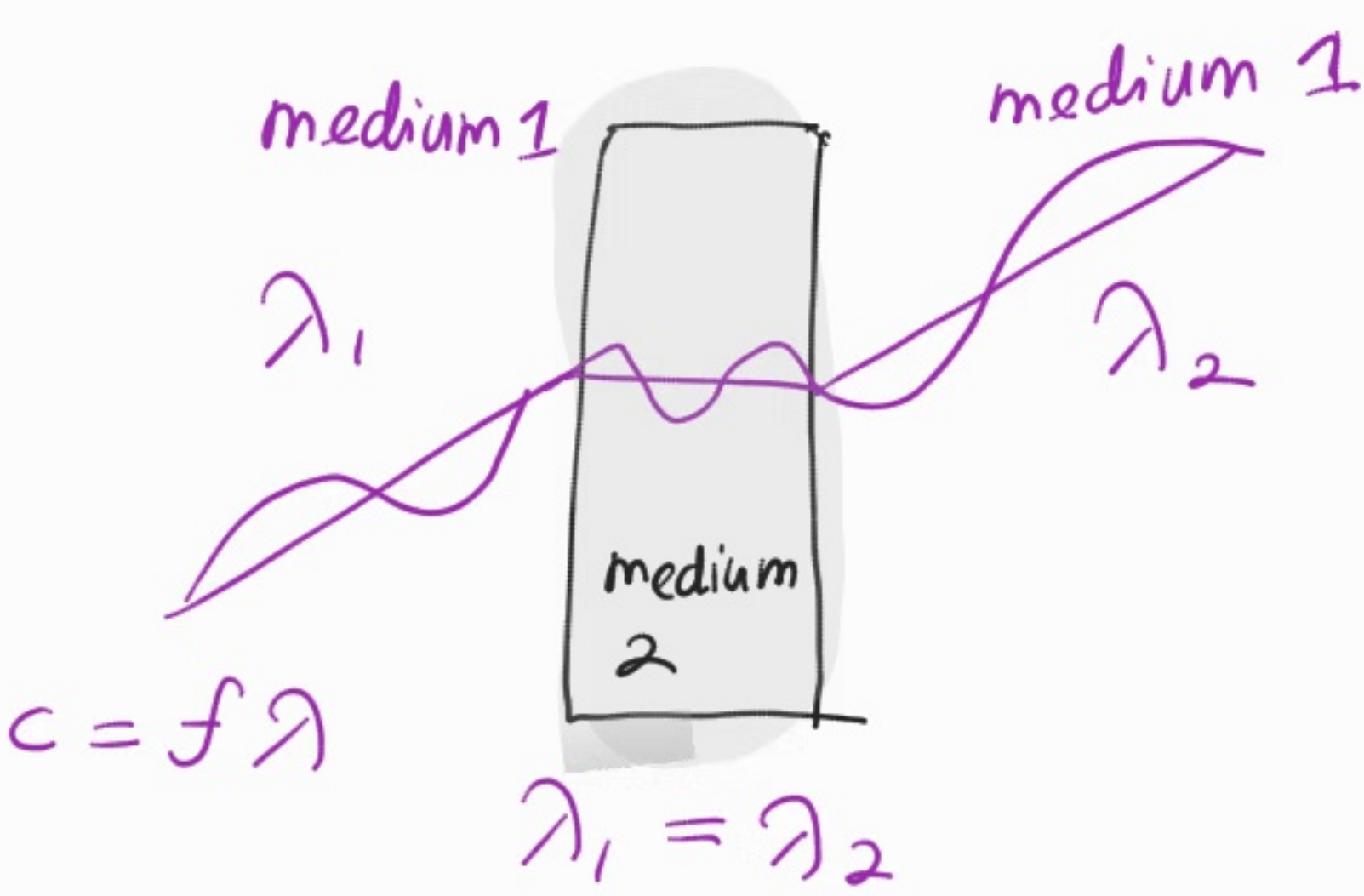
$$\boxed{\vec{P}_f = \hbar \vec{k}}$$

Compton Effect

- unusual results when X-rays are scattered on some materials.

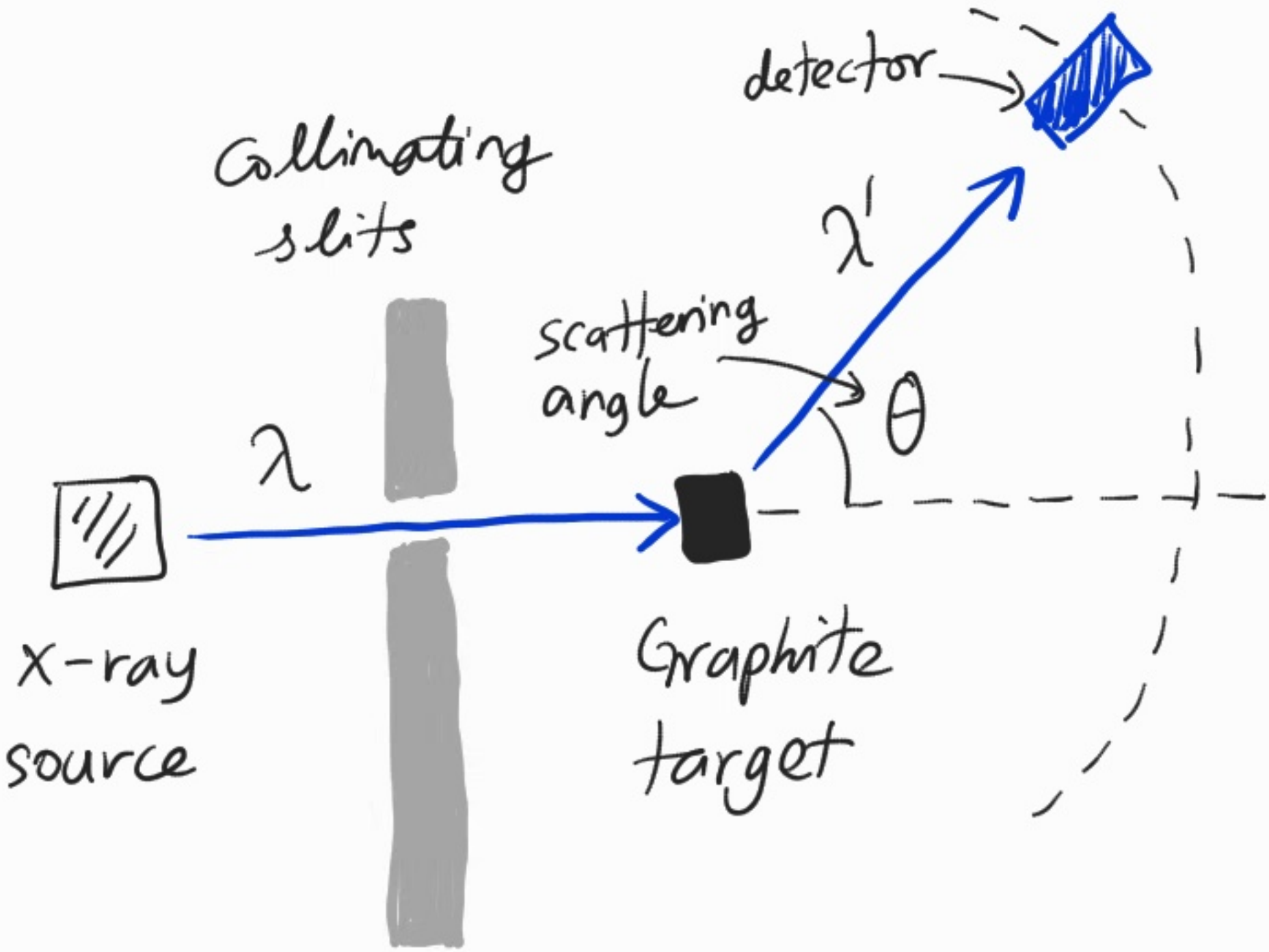
CLASSICAL THEORY.

→ when EM waves are scattered off some material (atoms), the wavelength of the incident & scattered should be the same.

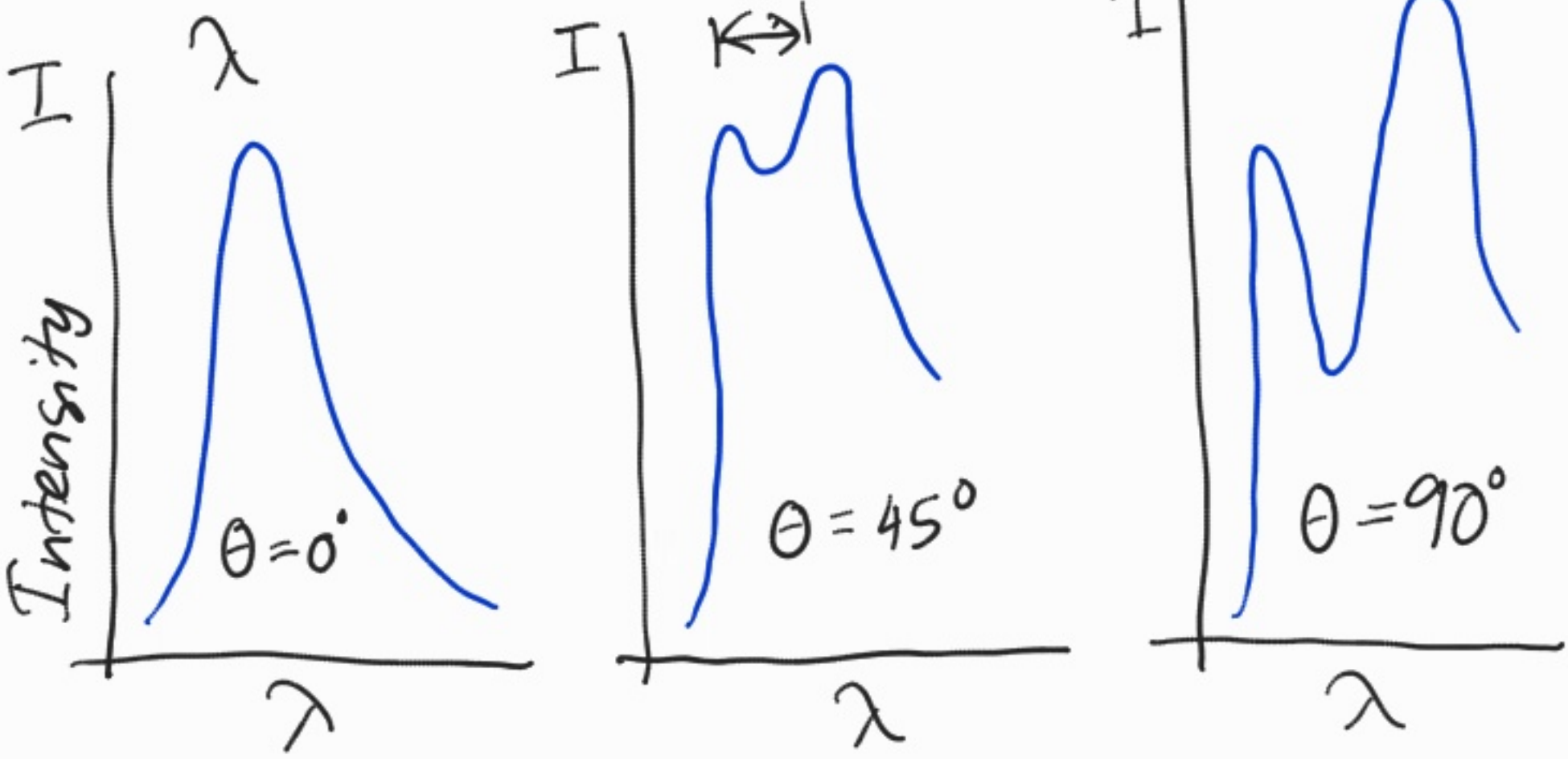


Compton Effect

- X-rays scattered on certain materials (like graphite) show a variation in λ as compared to the incident wave.



Compton Effect



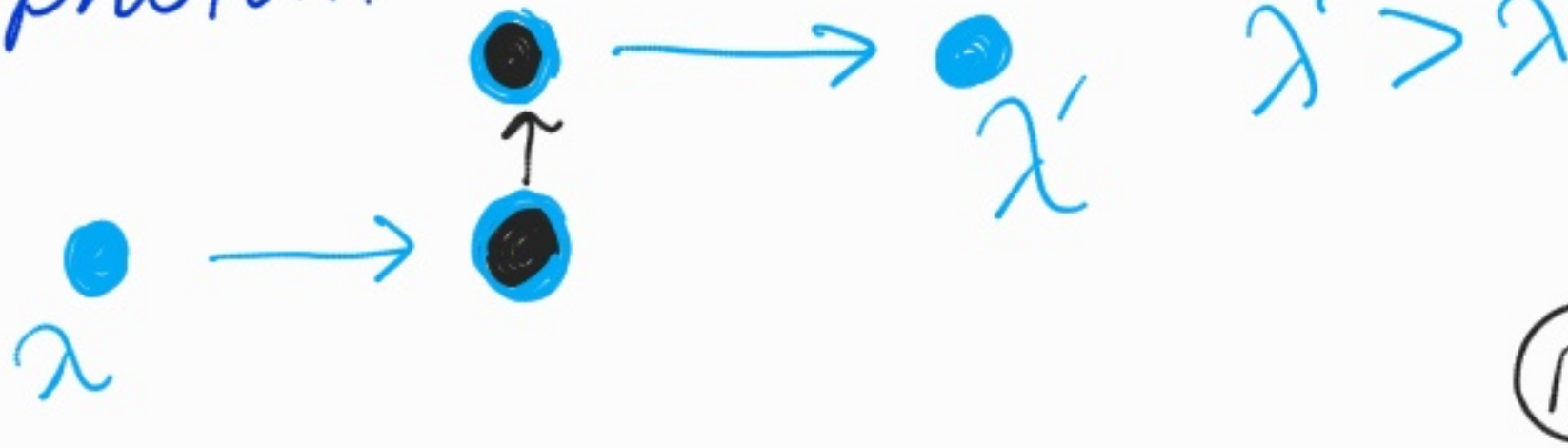
- Compton employed Einstein's idea of photons to explain the scattering results.
- EM waves could behave like a stream of photons. Thus solidifying quantum theory.

Compton Effect

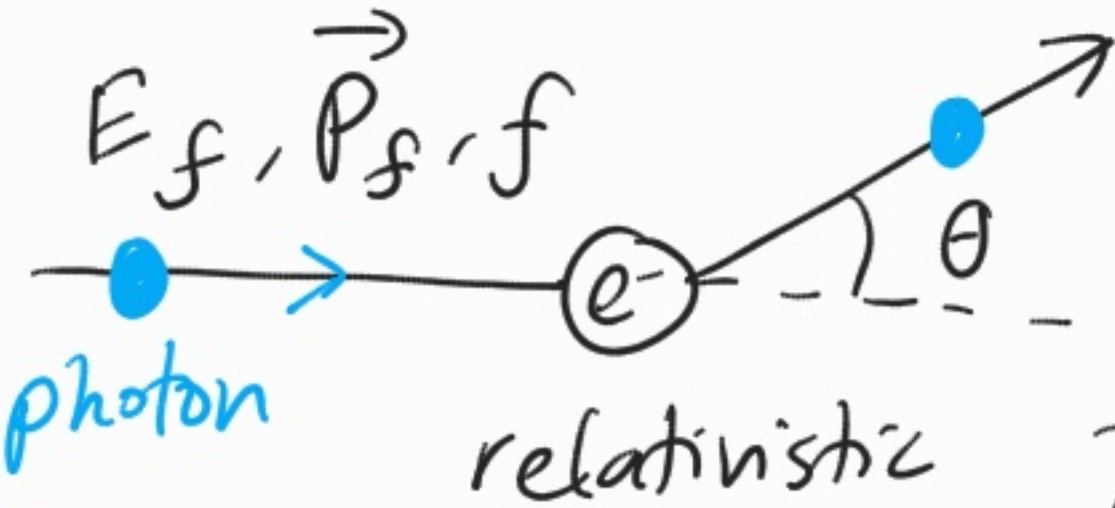
- The shift in wavelength in the Compton effect is called a Compton shift.

Compton's Explanation

- Incoming photon stream collides with a valence e⁻ in the graphite target.
- The incoming photon transfers some of its energy & momentum to the graphite electrons.
- Leaves the target as a scattered photon.



Compton Effect



relativistic electron at rest ($m_0 c^2$)

} after collision electron has $E, \vec{p}$

$\Rightarrow$ Dot Product for Scattering Angle:

$$\vec{P}_f \cdot \tilde{\vec{P}}_f = |\vec{P}_f||\tilde{\vec{P}}_f| \cos\theta$$

$\Rightarrow$ **Photon-Electron Energy Balance:**

$$E_f + m_0 c^2 = \tilde{E}_f + E$$

$\Rightarrow$ **Momentum Balance:**

$$\vec{P}_f = \tilde{\vec{P}}_f + \vec{p}$$

Compton Effect

$$\Rightarrow E_f + m_0 c^2 = \tilde{E}_f + E$$

$$\left[(E_f - \tilde{E}_f) + m_0 c^2\right]^2 = E^2$$

$$\Rightarrow E^2 = p^2 c^2 + m_0^2 c^4$$

$$(E_f - \tilde{E}_f)^2 + m_0^2 c^4 + 2 m_0 c^2 (E_f - \tilde{E}_f)$$

$$= p^2 c^2 + m_0^2 c^4$$

$$E_f^2 - 2\tilde{E}_f E_f + \tilde{E}_f^2 + m^0 c^4$$

$$+ 2 m_0 c^2 E_f - 2 m_0 c^2 \tilde{E}_f$$

$$= p^2 c^2 + m_0^2 c^4$$

Compton Effect.

$$\left\{E_f^2 - 2\tilde{E}_f E_f + \tilde{E}_f^2 + \cancel{m_0 c^4} + 2m_0 c^2 E_f - 2m_0 c^2 \tilde{E}_f\right\} = p^2 c^2 \cancel{+ m_0 c^4}$$

→ Divide by c^2.

$$\left\{\frac{E_f^2}{c^2} - \frac{2\tilde{E}_f E_f}{c^2} + \frac{\tilde{E}_f^2}{c^2} + 2m_0 \cdot E_f - 2m_0 \cdot \tilde{E}_f\right\} = p^2$$

$$\left[\frac{E_f}{c} - \frac{\tilde{E}_f}{c}\right]^2 + 2m_0 c\left[\frac{E_f}{c} - \frac{\tilde{E}_f}{c}\right]$$

$$= P_2$$

Compton Effect

→ Since $p_f = E_f / c$

$$(p_f - \tilde{p}_f)^2 + 2 m_0 c (p_f - \tilde{p}_f) = [p^2]$$

→ Vector form: $\vec{p}_f = \vec{\tilde{p}}_f + \vec{p}$

eliminate

$$\left[\vec{p}_f - \vec{\tilde{p}}_f\right]^2 = |\vec{p}|^2 = p^2$$

$$= \left[p_f^2 + \tilde{p}_f^2 - 2\vec{p}_f \cdot \vec{\tilde{p}}_f\right]$$

→ Since $\vec{p}_f \vec{\tilde{p}}_f = p_f \tilde{p}_f \cos\theta$

$$\left[p_f - \tilde{p}_f\right]^2 + 2 m_0 c \left[p_f - \tilde{p}_f\right]$$
$$= p_f^2 + \tilde{p}_f^2 - 2 \vec{p}_f \vec{\tilde{p}}_f$$

Compton Effect.

$$\cancel{p_f^2} + \cancel{\tilde{p}_f^2} - 2\tilde{p}_f p_f + 2 m_0 c p_f - 2 m_0 c \tilde{p}_f = \cancel{p_f^2} + \cancel{\tilde{p}_f^2} - 2 p_f \tilde{p}_f \cos\theta$$

$$-\tilde{p}_f p_f + m_0 c \left[p_f - \tilde{p}_f \right] = -p_f \tilde{p}_f \cos\theta$$

Divide by $\tilde{p}_f p_f$ & by $m_0 c$

$$-\frac{1}{m_0 c} + \frac{\cancel{m_0 c}}{\tilde{p}_f p_f}\left[p_f - \tilde{p}_f \right] = \frac{-\cos\theta}{m_0 c}$$

$$\left[\frac{1}{\tilde{p}_f} - \frac{1}{p_f} \right] = \frac{1}{m_0 c}\left[1 - \cos\theta \right]$$

Compton Effect.

Since $p_f = h/\lambda$

$$\frac{1}{\tilde{p}_f} - \frac{1}{p_f} = \frac{1}{m_0 c}(1-\cos\theta)$$

$$\therefore \quad \lambda' - \lambda = \frac{h}{m_0 c}[1-\cos\theta]$$

$$\lambda_c = \frac{h}{m_0 c} \Big\} \text{ Compton wavelength}$$

$$\lambda_c \sim 0.00243\,\text{nm}$$

$$\Delta\lambda = \lambda' - \lambda = \lambda_c(1-\cos\theta)$$

INELASTIC SCATTERING